

Research Objective

To explore the physiological mechanisms of human interactions in virtual and extended reality (VR/XR) environments, aiming to develop inclusive and accessible experiences that represent diverse user groups and promote equity in digital spaces.

Education

Carnegie Mellon University (CMU), Pittsburgh, PA	GPA: 3.77/4.00
Doctor of Philosophy in Mechanical Engineering	May 2026
New York Institute of Technology (NYIT), New York City, NY	GPA: 3.66/4.00
Master of Electrical and Computer Engineering	May 2021
Mercy College, Dobbs Ferry, NY	GPA: 3.71/4.00
Bachelor of Science, Biology	May 2018

Work Experience

Research Scientist Intern, Meta Inc. **2026**
User Experience and Robotics Systems

- Integrating physiological sensors to enable robotic systems to infer user intent and improve human-robot interaction.

Vision Pro Group (VPG), Apple Inc. **2025**
Vision Pro Ergonomics / Product Design Analyst Intern

- Conducted ergonomic assessments and data-driven analyses of Vision Pro hardware to optimize comfort, wearability, and user experience across diverse user populations.

Fellowships

- GEM Fellow, CMU, Mechanical Engineering, 2021 - Present
- McNair Scholar, Mercy College, Biology, 2018

Skills

Subject Expertise: Human-Computer Interaction, Extended Reality Interactions Development (Virtual Reality, Augmented Reality, and Mixed Reality interactions), Electromyography, Neuromechanics, Joint Kinematics, Character Animation, Physics-based Virtual Object Animation

Programming Language: Python, C#

Software: Unity, Vicon Nexus, Blender, WebXR, Unreal

Languages: Spanish (Native), English (near-native)

PhD Research

NeuroMechatronics Lab (Dr. Douglas Weber), Mechanical Engineering, CMU **2021**
Research Lead - Enabling Advanced Interactions through Closed-loop Control of Motor Unit Activity After Tetraplegia

- Validated Meta's system through direct collaboration, leveraging their myoelectric research prototype to drive forward digitized interactions and enhance accessibility solutions for individuals with spinal cord injuries.
- Created immersive 2D and 3D VR experiences, including games and tasks, with eye-tracking integration, allowing novel interactions with the prototype.
- Insights gained from real-time analysis of user performance with the myoelectric prototype were used to optimize interaction methods for VR and digitized applications.

Research Lead - Understanding the Neuromechanics of Object Manipulation in Physical and Virtual Environments

- Implemented Vicon markers for optical tracking of user kinematics and untethered Delsys sensors for real-time EMG readings to record muscle activities during interactions in both physical and virtual environments.
- Integrated kinematic and EMG data using the Neuromechanics framework to analyze changes in muscle activity and joint kinematics during digitized interactions.
- Worked to reveal how learning and embodiment in virtual environments affect cognitive and sensory integration, showing how the brain adapts to new tasks and interactions

Masters Research

Advised by N.Sertac Artan, Electrical and Computer Engineering, NYIT

2019

Research Lead - *Analysis of Virtual Reality and Myo Armband (EMG) Use in Prosthetic Rehabilitation*

- Conducted remote work using an Oculus Quest 2 and a Myo armband to develop and test a virtual arm with four controllable gestures (fist, open hand, wrist flexion, wrist extension) for user interaction.
- Tested the system with able-bodied participants, focusing on controlling the virtual arm, analyzing latencies between the Myo armband and the VR system, and minimizing these delays.
- Although the study was conducted with able-bodied participants, it suggested promising possibilities for improving device usability for amputees, to enhance gesture-based control and accessibility in virtual environments.

Graduate Coursework

In-Person Learning: Extended Reality interactions by the Human-Computer Interaction CMU program, Crafting Software, Machine Learning, Human Anatomy and Physiology, Neural Engineering Laboratory, Signals and Systems, Communication Theory

Online Learning (Coursera and edX): Principles of UX/UI Design, 3D Models in Virtual Reality, 3D Models for Virtual Reality, Interactive Computer Graphics, Developing AR/VR/MR/XR Apps with WebXR, Unity and Unreal

Posters

- Enabling Advanced Interactions through Closed-Loop Control of Motor Unit Activity After Tetraplegia, User Interface Software and Technology, PA, 2024
- Closed-loop Control of Motor Unit Activity After Tetraplegia for Human-Computer Interactions, Society for Neuroscience Annual Meeting, Chicago, 2024
- Realtime Detection and Decoding of Motor Unit Activity Using a Wearable Neuromotor Interface in a Person with Motor Complete Tetraplegia, Neural Control of Movement, Croatia, 2024
- Exploring the Neuromechanics of Feedback Influenced Object Manipulation in Virtual and Physical Environments, Society of Neuroscience, Washington DC, 2023
- Understanding the Neuromechanics of Haptic-Free Object Manipulation in Virtual and Physical Environments, Society of Neuroscience, CA, 2022

Outreach Experience

- Black History Month CMU MechE DEI VR Event, Spring 2024
- Biomechanics Day Outreach Event - Teaching High School and Middle School Students the Biomechanics of an Animated Character, Spring 2024
- Biomechanics Day Outreach Event - Teaching High School and Middle School Students the Biomechanics of Hand Rigging and Animation, Spring 2023
- MEGSO (Mechanical Engineering Graduate Student Association) Mentorship Chair, 2023-2024